

**Comprehensive
Geomechanical Evaluation
of Rock Mass
Characterization
Frameworks: RQD, RMR,
and the Q-System in
Global and Himalayan
Engineering Practice**

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Abstract

***Rock mass classification systems** form the foundational framework upon which modern underground engineering design is built. This document presents a comprehensive synthesis of the three most widely adopted **geomechanical characterization tools** — the **Rock Quality Designation (RQD)**, the **Rock Mass Rating (RMR) system**, and the **Tunnelling Quality Index (Q-system)** — tracing their theoretical foundations, parameter evaluation methodologies, and practical engineering applications. The **Geological Strength Index (GSI)** is further examined as a continuum-based extension of these systems, particularly suited for numerical modeling of weak and tectonically disturbed rock masses. Empirical methods for estimating the rock mass deformation modulus (E_m), including the Hoek-Diederichs sigmoid formulation, are reviewed and compared. Special attention is given to the complex geomechanical environment of the **Himalayan region**, where high in-situ stresses, thrust faulting regimes, and squeezing ground conditions impose significant challenges beyond the predictive capacity of standard empirical charts. Indian Standards (IS 13365) governing the application of these systems are outlined, and two landmark Indian case studies — the **Purulia Pumped Storage Project (West Bengal)** and the **Teesta Stage-III Hydroelectric Project (Sikkim)** — are examined to demonstrate real-world implementation. Emerging advancements including **deep learning-based RMR prediction** and the **Rock Mass Excavability (RME) Index** are also discussed. The document concludes that while these classification systems remain indispensable tools in geotechnical practice, their reliable application demands sound geological judgment, site-specific calibration, and supplementary numerical and instrumentation-based verification.*

Table of Contents

1. Introduction

2. The Foundation of Rock Quality Assessment : Rock Quality Designation(RQD)

2.1 Standardized Calculations and Core Logging Procedures

2.2 Indirect Estimation and Volumetric Joint Counts

3. The Geomechanics Classification : Rock Mass Rating(RMR)

3.1 Parameter Evaluation and Rating Assignments

3.1.1 Primary Geomechanical Parameters

3.1.2 The Role of Joint Orientation

3.2 Engineering Classifications and Meanings

4. The Tunnelling Quality Index : The Q-System

4.1 The Mechanics of the Q-Quotients

4.1.1 Quotient 1 : Block Size

4.1.2 Quotient 2 : Inter-Block Shear Strength

4.1.3 Quotient 3 : Active Stress

4.2 Support Design and the Equivalent Dimension(D_e)

5. Comparative Analysis of RMR and Q-Systems

5.1 Correlation Reliability and Site Specific Models

6. The Geological Strength Index(GSI) : The Continuum Approach

6.1 Qualitative Assessment and Mechanical Translations

7. Estimating The Rock Mass Deformation Modulus(E_m)

7.1 Comparative Review of Empirical Equations

8. In-Situ Stress Regimes : Mechanisms and Measurements

8.1 Vertical and Horizontal Stress Components

8.2 The Himalayan Tectonic Environment

9. Geomechanical Challenges : Squeezing Ground and Rock Bursts

9.1 Squeezing Assessment and Singh & Goel Criteria

10. Indian Standards and Case Study in the Subcontinent

10.1 IS 13365 : Quantitative Classification Guidelines

10.2 The Purulia Pumped Storage Project(PPSP)

10.3 The Teesta Stage-III Hydroelectric Project

11. Advancements in Characterization : AI and Digital Analytics

11.1 Deep Learning for RMR Prediction

11.2 Rock Mass Excavability (RME) Index

12. Conclusions and Engineering Significance

Works Cited

1. Introduction

The discipline of rock mechanics has undergone a profound transformation over the last century, evolving from a largely qualitative field based on empirical observations to a rigorous engineering science characterized by quantitative assessment and numerical modeling. Central to this evolution is the ability to characterize the rock mass—a complex, heterogeneous assemblage of intact rock blocks and intersecting discontinuities—in a manner that translates geological reality into engineering parameters. The **Rock Quality Designation (RQD)**, the **Rock Mass Rating (RMR)** system, and the **Tunnelling Quality Index (Q-system)** represent the three most influential pillars in this domain.¹ These systems allow engineers to systematically evaluate the competency of the ground, design support for underground excavations, and estimate the mechanical properties of materials that are far too large and complex to be tested in a laboratory environment.

2. The Foundation of Rock Quality Assessment: Rock Quality Designation (RQD)

The Rock Quality Designation (RQD) index, developed by **Don U. Deere** in 1964, was the first successful attempt to provide a quantitative index of rock mass quality based on drill core logging.¹ Before the advent of RQD, core recovery was the primary metric used to describe rock quality, but it often proved misleading. A rock mass could exhibit 100% recovery while being heavily fractured, thereby offering poor engineering conditions. RQD was specifically designed to rectify this by focusing on the "soundness" and frequency of natural discontinuities.⁴

2.1 Standardized Calculation and Core Logging Procedures

The calculation of RQD is mathematically straightforward but requires rigorous adherence to standardized procedures to ensure reproducibility and accuracy. It is defined as the percentage of intact rock core segments longer than 100 mm (4 inches) relative to the total length of the core run. The standard methodology, codified in **ASTM D6032**, emphasizes that only segments of "hard and sound" rock should be counted.⁴ This means that pieces which are highly weathered, soft, or showing signs of significant alteration are excluded from the summation, even if they meet the 100 mm length requirement.⁴

The reliability of RQD is inextricably linked to the quality of the drilling process. Mechanical breaks—fractures induced by the drilling action, core handling, or extraction—must be identified and fit back together, counting as a single continuous piece.⁴ These breaks are typically characterized by fresh, rough surfaces lacking the staining or mineralization found on natural joint surfaces.⁴ While RQD was originally optimized for **NX-size** core (54.7 mm diameter), contemporary practice allows for sizes ranging from **BQ to PQ**, provided that double-tube or triple-tube core barrels are utilized to minimize mechanical damage.⁴

RQD Percentage	Rock Quality Designation
0 - 25	Very Poor
25 - 50	Poor
50 - 75	Fair
75 - 90	Good
90 - 100	Excellent

The measurement procedure itself has been the subject of considerable debate, particularly regarding the orientation of measurements. The recommended practice is to measure the length of each core piece along its centerline or axis. Centerline measurements ensure that the RQD value is independent of the core diameter and avoids unduly penalizing the rock mass in cases where joints or fractures run parallel to the borehole.⁴

2.2 Indirect Estimation and Volumetric Joint Counts

In many geotechnical investigations, particularly during the early feasibility stages, borehole data may be limited. In such instances, engineers rely on surface mapping and scanline surveys to estimate RQD. A widely accepted method involves the use of the volumetric joint count (J_v), which is the sum of the number of joints per unit length for all joint sets present in the rock mass.⁹ For clay-free rock masses, the relationship is expressed as:

$$RQD = 115 - 3.3J_v$$

This empirical formula allows for the transition from three-dimensional outcrop observations to the one-dimensional index required by most classification systems. However, it is important to note that RQD is inherently sensitive to the orientation of the borehole or scanline relative to the dominant joint sets. A borehole drilled parallel to a major joint set may report an RQD of 100%, while a perpendicular borehole in the same rock mass might report zero.⁶ This directional bias is a significant limitation that practitioners must account for when using RQD as a stand-alone index or as an input for more complex systems.⁵

3. The Geomechanics Classification: Rock Mass Rating (RMR)

The Rock Mass Rating (RMR) system, developed by **Z.T. Bieniawski** between 1972 and 1973,

represents a significant leap forward in rock mass characterization. By integrating several significant geologic parameters into a single comprehensive index, RMR provides a method for incorporating the complex mechanics of in-situ rock into engineering design.² Since its inception, RMR has undergone multiple modifications, with the 1989 version (RMR89) remaining the global standard for civil engineering and mining applications.²

3.1 Parameter Evaluation and Rating Assignments

RMR is an additive system that assigns numerical ratings to six primary parameters. The sum of these ratings determines the final RMR value, which ranges from 0 to 100.²

3.1.1 Primary Geomechanical Parameters

1. **Uniaxial Compressive Strength (UCS) of Intact Rock:** This parameter evaluates the strength of the rock material itself. For high-strength rocks (UCS > 250 MPa), a maximum rating of 15 is assigned. As strength decreases, the rating drops, reflecting the increased susceptibility of the rock to fail under load.¹²
2. **Rock Quality Designation (RQD):** Directly incorporates the RQD index. A rock mass with RQD > 90% receives a rating of 20, while RQD < 25% receives only 3.¹⁴
3. **Spacing of Discontinuities:** This refers to the mean distance between adjacent fractures. Wider spacing (e.g., > 2 m) suggests a more massive and inherently stable rock mass, receiving a rating of 20. Closely spaced joints (< 60 mm) indicate a highly fractured mass and receive a rating of 5.¹²
4. **Condition of Discontinuities:** Often considered the most critical parameter, this evaluates the shear strength of the joints. It is a composite rating involving persistence (length), aperture (opening), roughness, infilling (gouge), and weathering.¹² For example, very rough, unweathered surfaces with no separation receive a maximum rating of 30, whereas continuous soft gouge more than 5 mm thick reduces the rating to zero.⁹
5. **Groundwater Conditions:** This assesses the presence of water and its pressure within the rock mass. Dry conditions receive 15 points, while flowing water, which can drastically reduce the effective stress and lubricate joint surfaces, receives a rating of 0.¹⁴

3.1.2 The Role of Joint Orientation

The sixth parameter in the RMR system is the "**Rating Adjustment for Discontinuity Orientations.**" Unlike the first five parameters, which are intrinsic properties of the rock mass, this parameter depends on the orientation of the engineering structure (tunnel, foundation, or slope) relative to the strike and dip of the dominant joint sets.²

Orientation Favorability	Tunnels (Adjustment)	Foundations (Adjustment)	Slopes (Adjustment)
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Very Favorable	0	0	0
Favorable	-2	-2	-5
Fair	-5	-7	-25
Unfavorable	-10	-15	-50
Very Unfavorable	-12	-25	-60

This adjustment factor is vital because it accounts for structural instability—such as wedge failures or toppling—that may occur regardless of the overall geomechanical quality of the rock mass. In tunnel design, for instance, joints striking parallel to the tunnel axis are generally considered unfavorable, as they facilitate the formation of large wedges in the roof.⁹

3.2 Engineering Classifications and Meaning

The final RMR value categorizes the rock mass into five classes, each with distinct engineering implications. These classes help engineers estimate the cohesion and friction angle of the rock mass, which are essential inputs for limit equilibrium and numerical analyses.¹²

Class No.	Quality	RMR Range	Cohesion (kPa)	Friction Angle (°)	Stand-up Time	Bolt Density Required
I	Very Good	81 - 100	> 400	> 45	10 years (15 m span)	No Support/ Spot Bolting
II	Good	61 - 80	300 - 400	35 - 45	6 months (8 m span)	0.6-0.7 bolt/m ²
III	Fair	41 - 60	200 - 300	25 - 35	1 week (5 m span)	1.0 bolt/m ²
IV	Poor	21 - 40	100 - 200	15 - 25	10 hours (2.5 m)	1.4-1.5 bolt/m ²

					span)	
V	Very Poor	< 20	< 100	< 15	30 minutes (1 m span)	Very Poor to even consider it for excavation

The concept of stand-up time, originally proposed by **Lauffer** (1958), is a cornerstone of RMR application in tunneling. It represents the period during which an unsupported excavation can remain stable. As the span of the tunnel increases, the stand-up time decreases exponentially, emphasizing the need for timely support installation in poorer rock classes.¹⁰

4. The Tunnelling Quality Index: The Q-System

While RMR is additive, the Q-system—developed by the **Norwegian Geotechnical Institute** (NGI) in 1974—is multiplicative. This structural difference makes the Q-system highly sensitive to individual parameters, particularly those related to shear strength and stress.³ The Q-value is calculated using six parameters grouped into three quotients, each representing a fundamental geomechanical attribute:

$$Q = \left(\frac{RQD}{J_n} \right) \times \left(\frac{J_r}{J_a} \right) \times \left(\frac{J_w}{SRF} \right)$$

4.1 The Mechanics of the Q quotients

4.1.1 Quotient 1: Block Size:- RQD/J_n

The first quotient uses RQD and the joint set number (J_n) to describe the structure of the rock mass. J_n values reflect the degree of jointing, ranging from 0.5 (massive rock) to 20 (crushed rock). This quotient provides a crude but effective measure of the relative block size within the mass.

4.1.2 Quotient 2: Inter-block Shear Strength:- J_r/J_a

The second quotient relates the joint roughness number (J_r) to the joint alteration number (J_a). This is perhaps the most significant part of the Q-system, as it represents the frictional and cohesive resistance of the discontinuities to shear. Rough, undulating joints with no alteration provide high strength, whereas slickensided joints with soft clay infilling offer minimal resistance to sliding.⁸

4.1.3 Quotient 3: Active Stress:- J_w/SRF

The third quotient integrates the environmental and stress conditions. J_w is the joint water reduction factor, accounting for the destabilizing effect of water pressure. The **Stress Reduction Factor (SRF)** is a sophisticated parameter that addresses loosening loads in shear zones, high-stress problems in massive rock (such as spalling or rockbursts), and squeezing or swelling behavior in incompetent rock.⁸

Q Value	Rock Quality Class
0.001 - 0.01	Exceptionally Poor
0.01 - 0.1	Extremely Poor
0.1 - 1	Very Poor
1 - 4	Poor
4 - 10	Fair
10 - 40	Good
40 - 100	Very Good
100 - 400	Extremely Good
400 - 1000	Exceptionally Good

4.2 Support Design and the Equivalent Dimension (D_e)

The Q-system is primarily a tool for permanent support design. The recommended support is determined by relating the **Q-value** to the "**Equivalent Dimension**" (D_e) of the opening.³

$$D_e = \frac{\text{Tunnel Span, Diameter, or Wall Height}}{\text{Excavation Support Ratio (ESR)}}$$

The ESR is an "importance factor" based on the purpose of the excavation and the required level of safety. For instance, temporary mine openings may have an ESR of 3.0 to 5.0, while critical nuclear power station caverns have an ESR of 0.5 to 0.8, reflecting a much lower

tolerance for deformation.⁸ The support chart, revised significantly in 1993 and 2002, provides specific recommendations for the thickness of fiber-reinforced shotcrete and the spacing of rock bolts based on the calculated Q and D_e .³

5. Comparative Analysis of RMR and Q-Systems

Because both systems aim to quantify the same geological reality, various researchers have attempted to establish linear or logarithmic correlations between them. The most famous correlation, proposed by **Bieniawski** in 1976, is:

$$RMR = 9 \ln Q + 44$$

However, subsequent experience has shown that these correlations are rarely universal and are highly sensitive to regional geology.¹⁸

5.1 Correlation Reliability and Site-Specific Models

The data used to derive early correlations were often widely scattered, and the 90% confidence limits were so large as to render the relationships of little practical value for detailed design.¹⁸

The divergence between the two systems arises from their different sensitivities: RMR explicitly includes intact rock strength and joint orientation, whereas the basic Q-system emphasizes the active stress state and inter-block shear strength.¹⁸

Region/Context	Proposed Correlation	Author
Worldwide (General)	$RMR = 9 \ln Q + 44$	Bieniawski (1976)
New Zealand (General)	$RMR = 5.9 \ln Q + 43$	Rutledge & Preston (1978)
Spain (General)	$RMR = 10.5 \ln Q + 43$	Abad et al. (1984)
South Korea (Faulted)	$RMR = 2.2 \ln Q + 22.4$	Recent Study ²¹
South Korea (Weak)	$RMR = 4.5 \ln Q + 40.9$	Recent Study ²¹
Turkey (Laminated)	$RMR = 7 \ln Q + 36$	Tugrul ²²

The sensitivity of the Q-system to parameter variations makes it a more effective tool for monitoring tunnel face conditions during construction, as it can capture subtle changes in joint alteration or stress that might not shift the RMR class. On the other hand, RMR is often favored

during the planning stages due to its simplicity and the relative ease with which its parameters can be estimated from core logging and preliminary investigations.²

6. The Geological Strength Index (GSI): A Continuum Approach

While RMR and Q-systems are excellent for designing tunnel support, they include parameters that are not always appropriate for numerical modeling—such as groundwater and orientation adjustments, which are better handled explicitly in finite element codes.²⁴ The Geological Strength Index (GSI) was developed by **Hoek and Brown** to provide a bridge between geological observations and the mechanical properties required for numerical simulation.²⁴

6.1 Qualitative Assessment and Mechanical Translation

GSI is fundamentally a qualitative system based on the visual examination of the rock mass structure (blockiness) and the surface condition of discontinuities.²⁴ It was specifically refined to handle very poor quality, tectonically disturbed rock masses where RQD becomes meaningless or difficult to measure.²⁴

The GSI chart allows an observer to estimate a value between 10 and 100. This "GSI number" is then used in conjunction with the UCS of the intact rock (σ_{ci}) and the petrographic constant (m_i) to calculate the strength parameters for the Hoek-Brown failure criterion.²⁴

$$GSI = RMR_{89} - 5 \text{ (for } GSI > 18 \text{)}$$

$$GSI = 9 \ln Q + 44 \text{ (for } GSI < 18 \text{)}$$

Note that for the RMR-GSI conversion, the groundwater rating is set to 15 (dry) and the orientation adjustment to 0, as these factors are accounted for elsewhere in the numerical model.²⁸ GSI allows for the rock mass to be treated as a mechanical continuum, making it the preferred system for estimating the strength and deformation properties of jointed rock masses in complex engineering problems.²⁴

† Author's Note: A noteworthy mathematical observation arises from the juxtaposition of two expressions in this document: the worldwide RMR-Q correlation proposed by Bieniawski (1976), $RMR = 9 \ln Q + 44$, and the GSI-Q relationship for poor quality rock masses, $GSI = 9 \ln Q + 44$ (for $GSI < 18$). Since both expressions are algebraically identical, it follows that $GSI = RMR$ in this domain. However, this equivalence is not coincidental — GSI was itself derived from RMR_{89} through the relationship $GSI = RMR_{89} - 5$ (for $GSI > 18$). For very poor quality, tectonically

disturbed rock masses where GSI falls below 18, the RMR system loses reliability as a characterization tool. Hoek and Brown therefore substituted the Q-based expression for this lower range, which inherently mirrors the Bieniawski worldwide correlation. The mathematical equivalence observed here is thus a transition artifact reflecting the shared empirical foundations of these classification systems, rather than an indication that RMR and GSI are universally interchangeable.

7. Estimating the Rock Mass Deformation Modulus (E_m)

The deformation modulus is a critical parameter for calculating the settlement of dam foundations and the convergence of tunnels. Because in-situ tests—such as plate loading, dilatometer, and flat jack tests—are exceedingly expensive and logistically difficult, empirical estimation from classification indices is common practice.²⁹

7.1 Comparative Review of Empirical Equations

Numerous authors have proposed equations relating E_m to RMR, Q, and GSI. These can be broadly divided into non-normalized equations (predicting E_m directly) and normalized equations (predicting the ratio of rock mass modulus to intact rock modulus, E_m/E_i).³⁰

Author	Equation	Application
Bieniawski (1978)	$E_m = 2RMR - 100$	RMR > 50
Serafim & Pereira (1983)	$E_m = 10^{(RMR-10)/40}$	RMR ≤ 56
Barton (2000)	$E_m = 10Q_c^{1/3}$ where $Q_c = Q \cdot (\sigma_c/100)$	General
Hoek & Diederichs (2006)	$E_m = E_i \{ 0.02 + [(2-D) / (2 + 2e^{(60+15D-GSI)/11})] \}$	General, with Disturbance

The **Hoek-Diederichs equation** is particularly significant as it incorporates a "**Disturbance Factor**" (D), which accounts for the reduction in modulus due to blast damage and stress relaxation around an excavation. This sigmoid function ensures that the modulus does not increase indefinitely as the rock mass becomes more massive, providing a more realistic upper bound based on the intact rock modulus (E_i).²

8. In-Situ Stress Regimes: Mechanisms and Measurement

Knowledge of the pre-excavation state of stress is a prerequisite for the rational design of large underground openings. Rock at depth is subjected to stresses from the weight of the overlying strata (overburden) and tectonic forces.³⁴

8.1 Vertical and Horizontal Stress Components

The vertical stress (σ_v) is typically estimated as the product of the depth (z) and the unit weight of the rock mass (γ):

$$\sigma_v = \gamma z$$

A standard rule of thumb is that 40 meters of rock overburden induces approximately 1 MPa of vertical stress.³⁴ Horizontal stresses (σ_h), however, are much more difficult to predict. The ratio of average horizontal stress to vertical stress is denoted by k .

$$k = \frac{\sigma_h}{\sigma_v}$$

In tectonically active regions, such as the Himalayas, k can be significantly greater than 1.0, particularly at shallow depths. This is often attributed to the "locked-in" stresses from crustal plate movements or the effects of erosion and deglaciation.³⁶

8.2 The Himalayan Tectonic Environment

The **Eastern Himalayan Syntaxis (EHS)** exhibits a complex stress regime driven by the Indian-Eurasian plate collision. Recent hydraulic fracturing tests in this region have revealed that both maximum (σ_H) and minimum (σ_h) horizontal principal stresses increase linearly with depth. At depths less than 300 meters, a thrust faulting stress regime ($\sigma_H > \sigma_h > \sigma_v$) is prevalent. As depth increases, the regime often evolves toward strike-slip faulting.³⁷

The orientation of these stresses is often governed by regional tectonic trends but can be locally rotated by major fault systems. For instance, in some Himalayan sub-regions, the maximum horizontal stress is oriented NEE, reflecting the prevailing convergence of the plates.³⁷ Understanding these stresses is critical because high differential stresses can lead to structural failure even in rock masses that classification systems might otherwise categorize as "good" quality.³⁴

9. Geomechanical Challenges: Squeezing Ground

and Rockbursts

Himalayan tunneling projects frequently encounter "squeezing" ground—a time-dependent, large-scale deformation of the tunnel peripheries that occurs when the induced stress exceeds the strength of the rock mass.⁴¹

9.1 Squeezing Assessment and Singh & Goel Criteria

Singh and Goel (2011) established a quantitative scale for squeezing based on the percentage of tunnel closure (radial convergence). This scale is vital for adapting support systems in extremely weak Himalayan rocks, such as phyllites and graphitic schists.⁴¹

Squeezing Severity	Tunnel Closure (%)
Very Mild	< 2
Mild	2 - 3
Mild to Moderate	3 - 4
Moderate	4 - 5
High	5 - 7
Very High	> 7

Projects such as the **Sanjen Hydroelectric Project** have found that standard empirical support charts from the Q-system are often inadequate for high-squeezing sections. In these cases, support must be augmented with steel ribs, lattice girders, and umbrella forepoling to reinforce the rock mass ahead of the tunnel face.⁴¹

10. Indian Standards and Case Studies in the Subcontinent

India has a long history of utilizing geomechanical classifications in its massive hydropower development program. The **Bureau of Indian Standards (BIS)** has codified these systems into the IS 13365 series of codes.⁴³

10.1 IS 13365: Quantitative Classification Guidelines

- **Part 1: RMR for Engineering Properties:** Provides the procedure for RMR determination

and its use in estimating stand-up time and unsupported spans.⁴⁴

- **Part 2: Q-system for Support Pressure:** Outlines the calculation of rock mass quality (**Q**) for predicting support pressure in tunnels.⁴³
- **Part 3: Slope Mass Rating (SMR):** Adapts the RMR system for rock slope stability analysis.
- **Part 4: GSI for Rock Strength:** Covers the methodology for qualitative GSI assessment and its translation into strength parameters for numerical analysis.²⁸

10.2 The Purulia Pumped Storage Project (PPSP)

The **Purulia Pumped Storage Project in West Bengal** is a landmark in Indian geotechnical engineering. This 900 MW project involved the construction of large underground caverns and rock-fill dams in a complex geological setting.⁴⁶

Feature	Specification
Installed Capacity	4 x 225 MW
Powerhouse Cavern	157 m (L) x 22.5 m (W) x 48.7 m (H)
Transformer Cavern	119 m (L) x 17 m (W) x 17 m (H)
Maximum Head	177 m
Reservoir Storage	13 million m^3 each (Upper/Lower)

The rock mass was characterized using RMR and Q-systems, with input parameters derived from exploratory drifts and 3D geological mapping.²³ Given the massive scale of the caverns, the projects utilized extensive instrumentation—including borehole extensometers, piezometers, and load cells—to monitor the stability of the large spans during excavation. This monitoring verified the adequacy of the support designs and provided a critical safety check in the heterogeneous rock of the Ajodhya Hills.⁵⁰

10.3 The Teesta Stage-III Hydroelectric Project

Located in **Sikkim**, the **Teesta Stage-III project** features one of the longest headrace tunnels in the Himalayas. The tunnel passes through quartz-mica schist and gneiss, frequently intersecting major shear zones and faults. Support design was conducted using a multi-parameter approach, utilizing RMR, Q, and the Rock Mass Index (RMI).⁵¹ **Numerical modeling** using Phase2 software was employed to quantify the plastic zone around the tunnel in squeezing sections, demonstrating that while empirical systems provided a baseline,

site-specific analytical verification was required for the most challenging ground conditions.⁵¹

11. Advancements in Characterization: AI and Digital Analytics

The traditional application of RMR and Q-systems is often criticized for its subjectivity—different observers may assign different ratings to the same rock mass. Recent advancements in machine learning (ML) and computer vision are addressing this challenge.¹¹

11.1 Deep Learning for RMR Prediction

Research has shown that deep learning models, such as **Artificial Neural Networks (ANN)**, can accurately predict RMR values using only the six parameters of the Q-system as inputs. By training these models on large datasets of independently determined field measurements, engineers can achieve R^2 values higher than **0.99**. This eliminates the need for separate fieldwork for both systems, allowing for a more efficient and objective classification process.¹¹

11.2 Rock Mass Excavability (RME) Index

As TBM tunneling becomes more prevalent, the need for an excavability-focused classification has grown. The **Rock Mass Excavability (RME)** index, developed by Bieniawski and Celada (2004), provides a tool specifically for predicting TBM performance. It considers the rock's uniaxial compressive strength, drillability/abrasivity, and stand-up time to predict the average rate of advance (m/day), demonstrating the ongoing specialization of rock mass classification systems to meet modern engineering needs.⁵²

12. Conclusions and Engineering Significance

The triad of RQD, RMR, and the Q-system remains the cornerstone of modern rock engineering. RQD, despite its one-dimensional nature, provides the essential starting point for any geomechanical assessment, emphasizing the importance of high-quality core recovery and sound rock identification. RMR, with its additive logic and focus on structural orientation, serves as a robust tool for general engineering planning and stand-up time estimation. The Q-system, through its logarithmic sensitivity and stress-aware **SRF** parameter, remains the premier tool for detailed tunnel support design in complex environments.

The development of the GSI system has successfully extended the reach of rock mass characterization into the realm of numerical modeling, allowing for the rational analysis of very poor quality and disintegrated rock. Meanwhile, the extreme conditions of the Himalayan region have driven the evolution of site-specific criteria for squeezing ground and the formalization of Indian Standards that integrate these global tools into a regional context.

As the industry moves toward digital transformation, the integration of AI and automated digital mapping promises to enhance the objectivity and precision of these systems. However, the fundamental principle remains: rock mass classifications are not a substitute for thorough

geological investigation but a means to organize and quantify that investigation for engineering design. The successful completion of projects like Purulia and Teesta demonstrates that when applied with engineering judgment and supplemented by monitoring and numerical analysis, these geomechanical frameworks provide a reliable pathway for designing and building in the earth's most challenging environments.

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